

Post-Quantum Cryptography

for the Novice, the Enjoyer, the Deployer and the Academic

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BalCCon 2026, Novi Sad

Cryptography is already a part of your life



messages



payments



your car



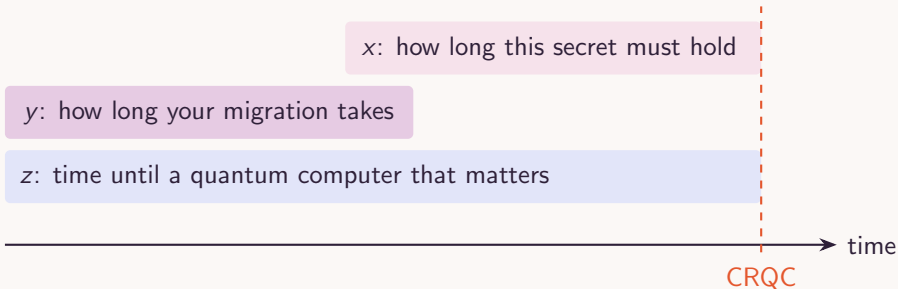
your ID



every connection

- All of the cryptography is being replaced in the next ten years.
- Touches a lot of code.
- Math is too complicated for me
It's a skillset, not a separate function.

Mosca's inequality: How long until my secret is revealed?



If $x + y > z$, you are already too late.

M. Mosca. *Cybersecurity in an Era with Quantum Computers: Will We Be Ready?* IEEE Security & Privacy 16(5), 2018.

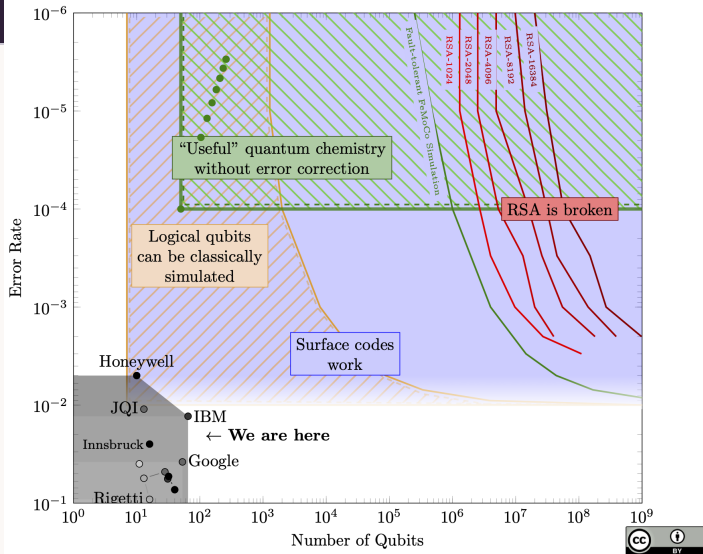


Figure 1: https://sam-jaques.appspot.com/quantum_landscape

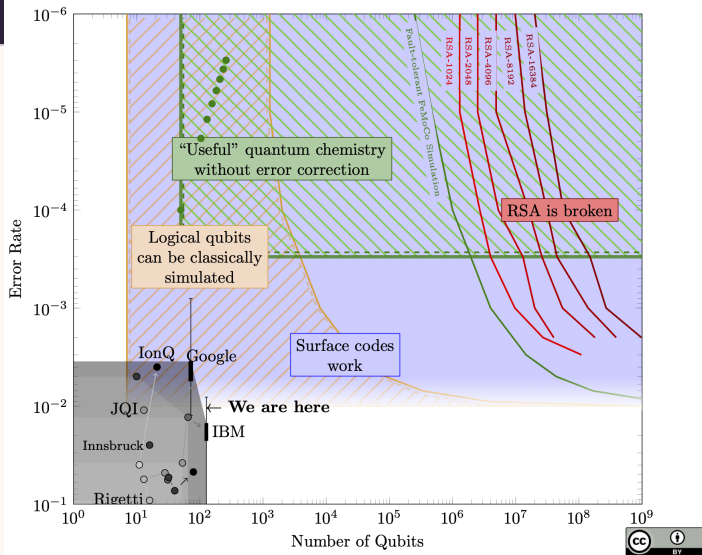


Figure 2: https://sam-jaques.appspot.com/quantum_landscape

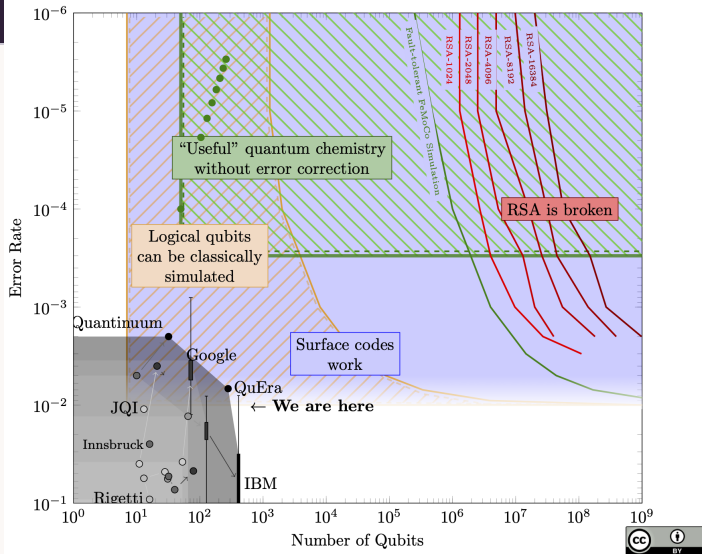
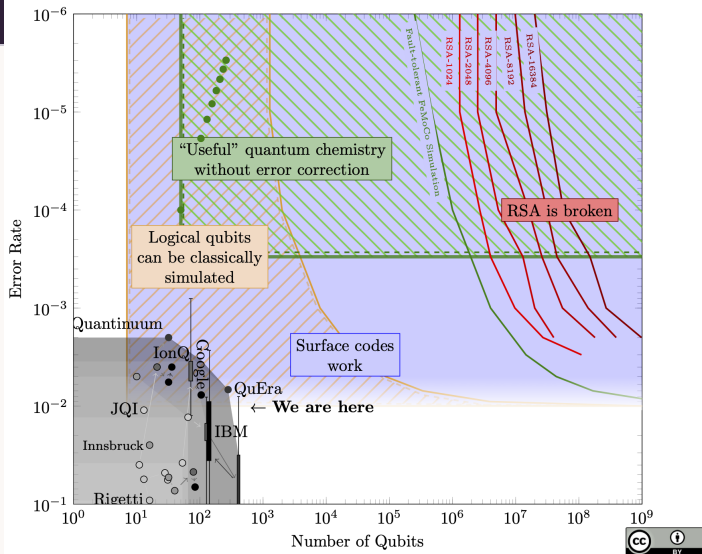
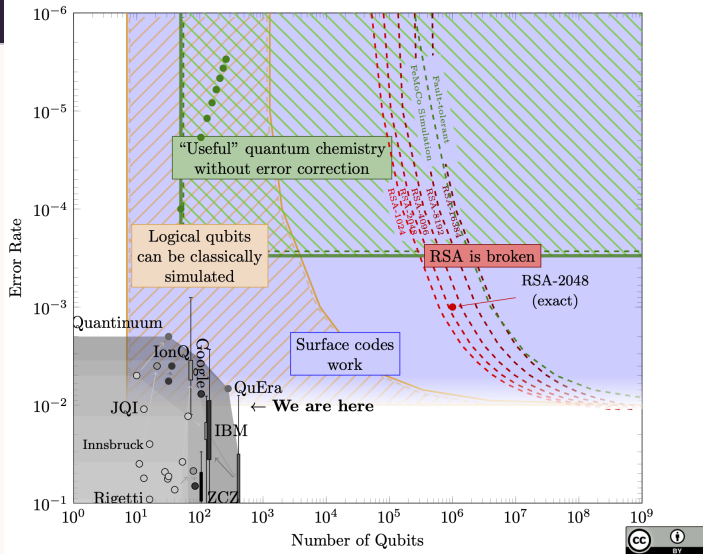


Figure 3: https://sam-jaques.appspot.com/quantum_landscape

Figure 4: https://sam-jaques.appspot.com/quantum_landscape

Figure 5: https://sam-jaques.appspot.com/quantum_landscape

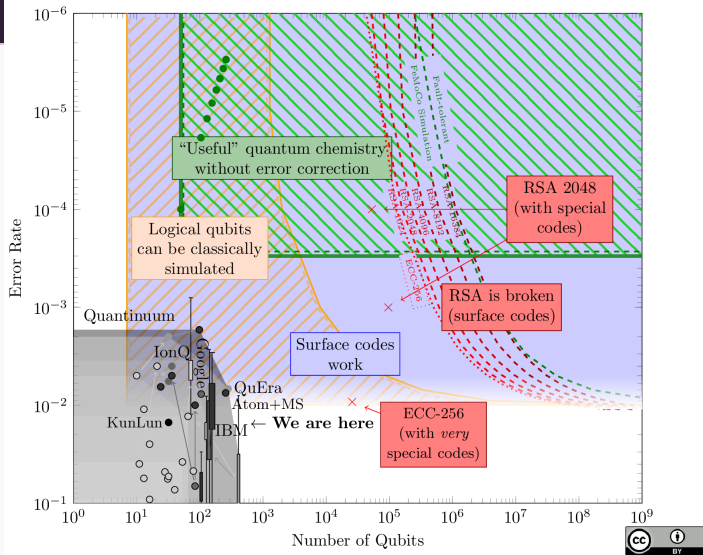


Figure 6: https://sam-jaques.appspot.com/quantum_landscape

When is a scheme secure?



- When has enough time passed?
- Reductionist security:
show breaking the scheme is as hard as breaking underlying assumption
- Reason about model, tightness of reduction, which problem.

Security Levels

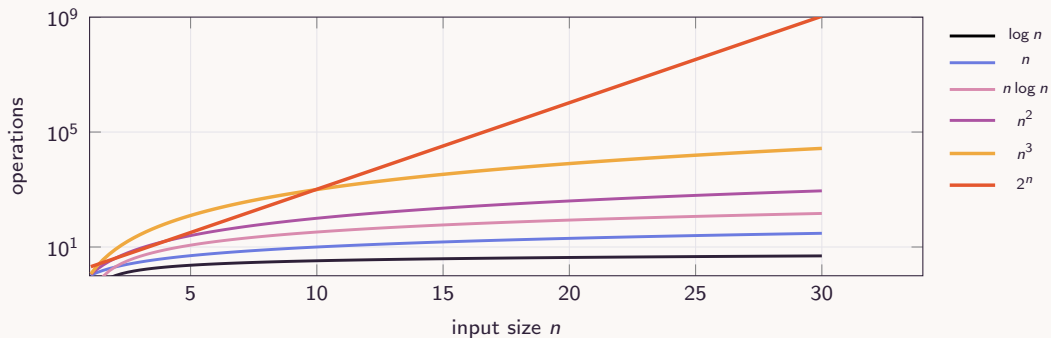
NIST level	at least as hard as
1	recovering an AES-128 key
2	finding a SHA-256 collision
3	recovering an AES-192 key
4	finding a SHA-384 collision
5	recovering an AES-256 key

- “ k -bit security” means roughly 2^k operations: 128 bits is AES-128, RSA-3072 and P-256.
- Rough estimate, independent of hardware, memory, target...
- **Quantum cost models:** the qubit count for RSA-2048 fell by a factor of twenty.

How to factor 2048 bit RSA integers with less than a million noisy qubits, Craig Gidney,

<https://arxiv.org/pdf/2505.15917>

Just double the parameters



Also doubles the work you need to do!
Terabytes of RSA keys

Why am I qualified to talk about this?

- PhD Student in $^*:-^*$ Privacy in a post-quantum world $^*_-^*$
- PQ protocols, primitives and assumptions
- deeply optimistic



I also did applied crypto last two summers



[Privacy](#) [Terms](#)

Talk touches a lot of points. If you ever zone out, just zone back in later.

When to switch to PQ?

Shared secrets are fine. AES, SHA-2, HMAC: Grover's algorithm.

Asymmetric is vulnerable. RSA, Diffie-Hellman, ECDSA: Shor's Algorithm.

Encryption

Harvest-now-decrypt-later.

deadline: the day the data is created

Signatures

Forgery with a CRQC.

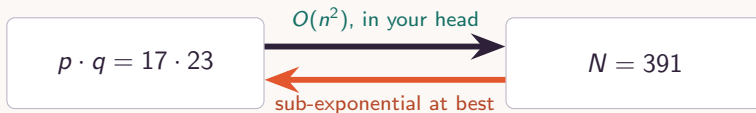
deadline: the day you stop accepting it



the Novice

“Why does a quantum computer break cryptography?”

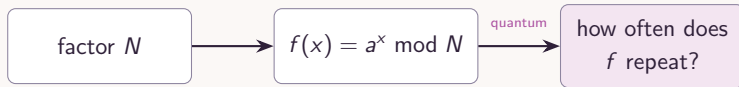
A hard problem becomes easy with a friend's secret



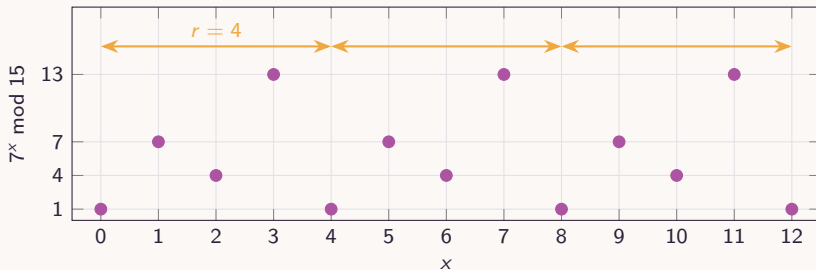
- Polynomial complexity for multiplication.
- general number field sieve runs in $\exp(O((\log N)^{1/3}(\log \log N)^{2/3}))$.
- NP-hard: easy to verify, hard to search.

That bounds the *worst* instance, and cryptography needs the *average* instance to be hard.

Shor reduces factoring to frequency finding

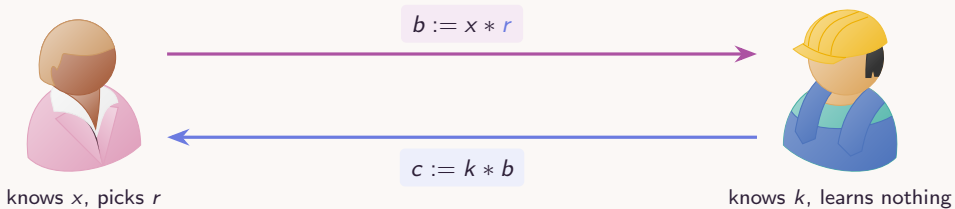


P. Shor, 1994/1997.



RSA security depends on the attacker not knowing the order of the group.

Blinding operations



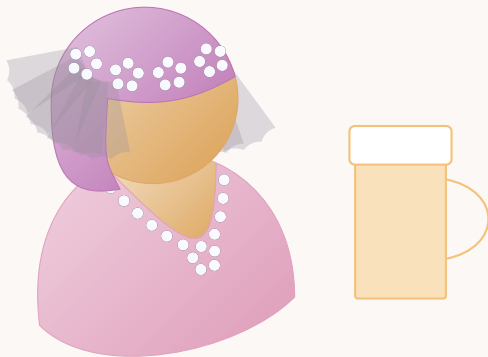
Hidden shifts in commutative settings: Kuperberg



- Same commutative structure, secret moved from *period* to *offset*.
- Then quantum computers have **subexponential** advantage.

G. Kuperberg. A subexponential-time quantum algorithm for the dihedral hidden subgroup problem.
SIAM J. Comput. 35(1), 2005.

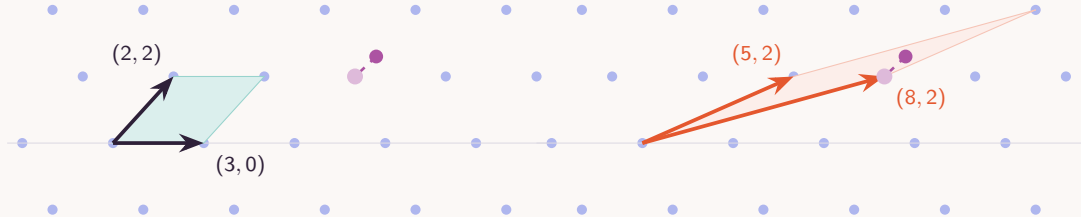
Remove the frequency.



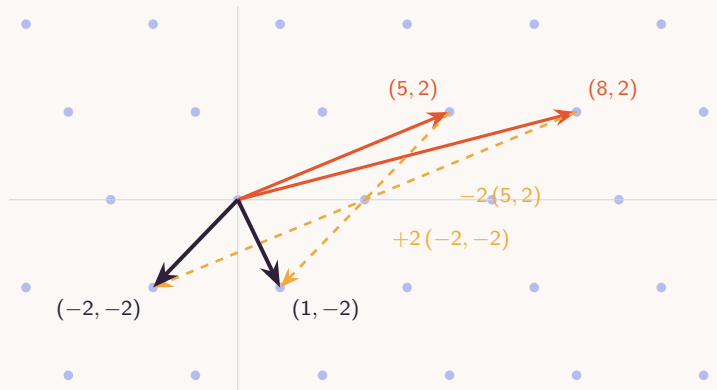
the Enjoyer

“What Alice did last night.”

The same lattice, described twice



All about that base



repeat

$$\mu \leftarrow \left\lceil \frac{\langle b_1, b_2 \rangle}{\langle b_1, b_1 \rangle} \right\rceil$$

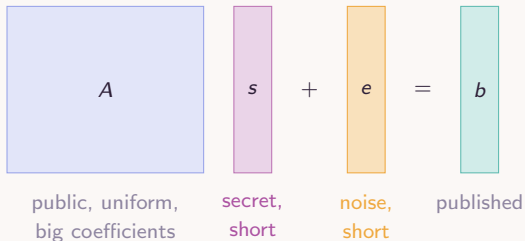
$$b_2 \leftarrow b_2 - \mu b_1$$

swap if $\|b_2\| < \|b_1\|$

until no swap

- Subtract the nearest multiple, swap, repeat. Replace the inner products with division and this is Euclid's algorithm.
- LLL is the approximate version.

Lattices: linear algebra with some noise



The diagram illustrates the equation $As = b + e$ using colored boxes and text labels. A large light blue square labeled A is on the left. To its right is a tall, narrow purple rectangle labeled s . Further right is a plus sign, followed by a tall, narrow orange rectangle labeled e , then an equals sign, and finally a tall, narrow teal rectangle labeled b . Below each box is a text label: 'public, uniform, big coefficients' under A , 'secret, short' under s , 'noise, short' under e , and 'published' under b .

$$A s + e = b$$

public, uniform, big coefficients secret, short noise, short published

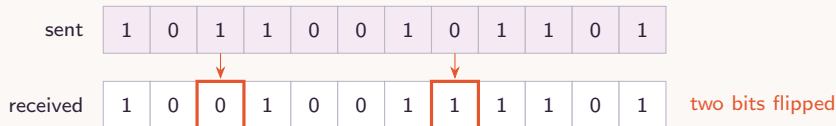
- Given A and b , find s .
Drop e and this is Gaussian elimination.
- With e , you need lattice reduction.
- Everything else is optimization:
rings and modules to shrink the key, rounding for determinism.

A well-studied problem

- system of univariate linear equations over integers with some noise
- Number fields, p -adic numbers, Abelian varieties:... can be restated as a lattice efficient general lattice reduction solve far more than cryptography.
- lattices have been studied as long as elliptic curves, but patents limited use
- Lattice reduction started as an attack broke knapsack cryptosystems in the 80's

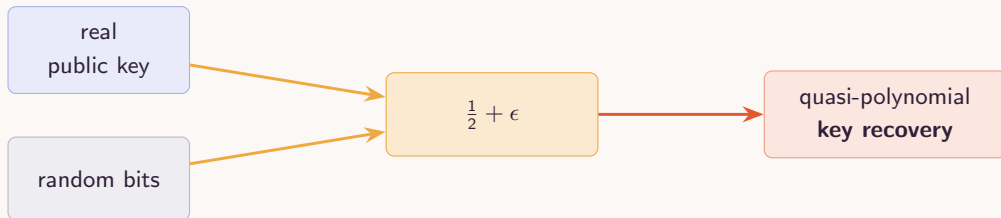
Rise from the ashes of cryptanalysis.

Codes: the same equations, counted differently



- Underdetermined linear system.
Sparse solution instead of a small one.
- The secret is an efficient decoder.
- Public: the same code, scrambled until it looks random.
- McEliece, 1978. Older than RSA is old.
- ~~conservative choice~~. Distinguisher attack.

What a distinguisher actually means

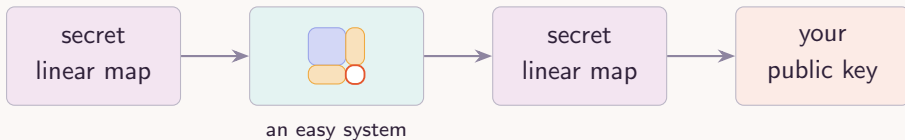


- Why care if someone can tell your public key from noise?
- The key has structure nobody knew about.
- Structure enables cryptanalysis.
- The distinguisher became a key recovery in a few weeks.
- BIKE and HQC are untouched.

Quasipolynomial Cryptanalysis of the McEliece Cryptosystem (or: PIR Meets McEliece)

Ghoshal, Ishai, Jain, Sun, ia.cr/2026/1630

Multivariate: hide an easy system inside a hard-looking one



- solving multivariate, random quadratic equations over a finite field
- every equation holds exactly, structure is hidden in linear maps
- 96-byte signatures, public keys in the tens of kilobytes

The easy system: oil and vinegar

vinegar	oil
$v \times v$	$v \times o$
$o \times v$	0

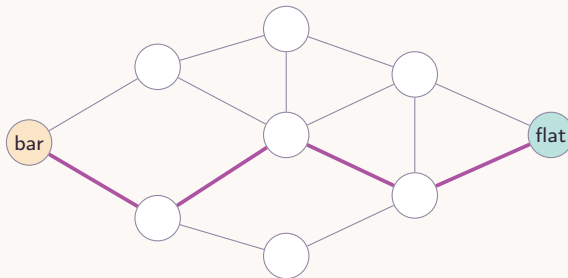
no oil times oil: fix
the vinegar and the
rest is *linear*

sig: 96 bytes

pk: 10s kB

- Signing: guess the vinegar, solve the rest linearly. Forging without the secret maps: find hidden oil subspace.
- **Balanced** oil and vinegar died in 1998 (Kipnis-Shamir). **Unbalanced**, 2 – 3× more vinegar, still standing
- Shrinking the public key with more structure is risky

Alice walks home alone at night



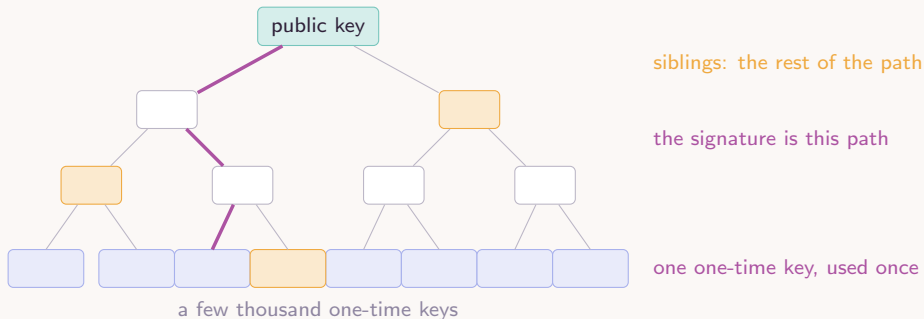
- Start: Bar, End: Got home safe text
- Nobody can reconstruct the path.
- Nodes are elliptic curves, steps are maps between them: Isogeny walk!

Isogenies: small, elegant, and a lot of work

- SQIsign: 148-byte signatures
Public key and signature together fit in a tweet.
- Signing is slow, constant-time implementation is still open, and the security assumption is the youngest of the lot.
- CSIDH is vulnerable to Kuperberg
- SIDH, the previous favourite, was broken in an afternoon in 2022.

You young people with your fancy math!
We know how to build cryptography.

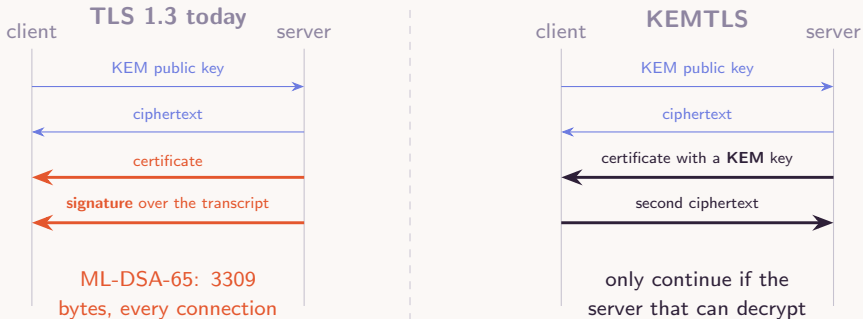
Symmetric: One way to signatures



- A hash turns a secret into a one-time key.
- Clunky: either keep state and never reuse a leaf, or go stateless and large
- Security reduces to underlying hash function. We need one anyways, somewhere

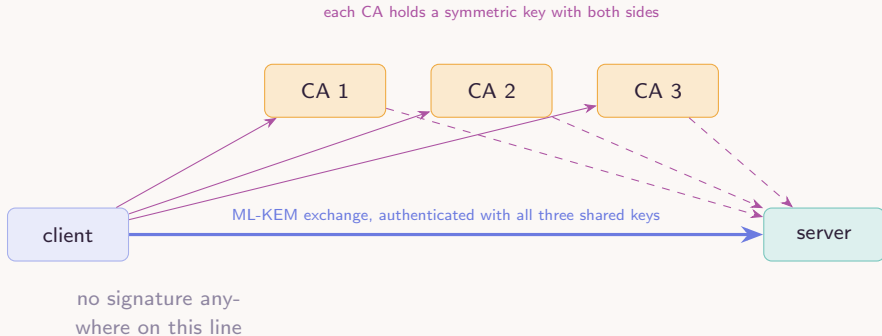
But signatures are expensive!

KEMTLS: authenticate via decryption



- No handshake signature, decryption is proof of private key possession.
- **implicit** authentication, one extra round.
- The CA still signs your certificate. KEMTLS removes the one signature, MTCs remove the four made in advance.

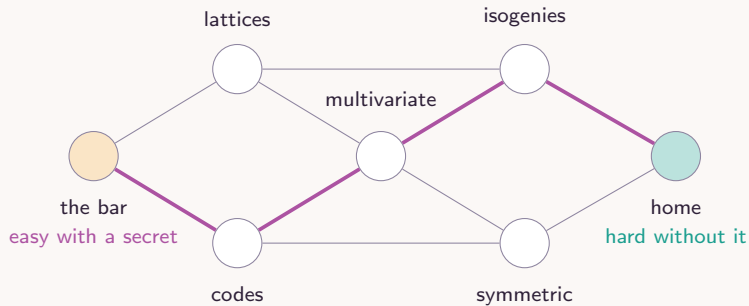
Down the symmetric rabbithole



- The server's DNS carries a key per CA, encrypted to that CA and bound to the hostname. The client asks three CAs to pass a fresh key through. Symmetric throughout, already post-quantum.
- **Three CAs must collude** for an attack.
- Every CA learns which sites you visit, live. Like OCSP, but worse.

Adam Langley, <https://www.imperialviolet.org/2024/04/07/letskerberos.html>

Her walk, in hindsight



Where do we go?



the Deployer

“The ethical stage: committing to an algorithm”

Start where you are

standard	what it is	status
ML-KEM (FIPS 203)	key exchange, lattices	shipping, everywhere
ML-DSA (FIPS 204)	signatures, lattices	the default
SLH-DSA (FIPS 205)	signatures, hashes	slow, conservative
LMS / XMSS (SP 800-208)	signatures, hashes, stateful	firmware
FN-DSA (FIPS 206)	signatures, lattices, small	still draft
HQC	key exchange, codes	backup, picked 2025

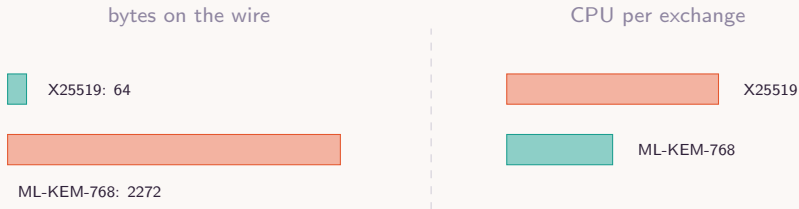
- That's it.
- Everything else is either research prototype or in standardization process.
- You go to war with the algorithms you have, not the ones you wish you had.

<https://educatedguesswork.org/posts/pq-emergency/>, Eric Rescola

Nobody asked you

- The internet has decided. Browsers, OpenSSL, Go, Apple, the big CDNs: post-quantum key exchange is on by default and the share only goes up.
- the question is *when* rather than *whether*.
- Betting against engineering problems usually means losing.

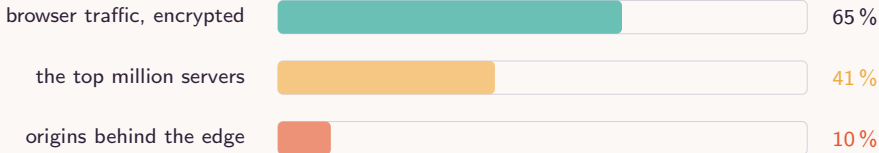
But it is bigger



- Lattices are bigger, but **faster** than elliptic curves.
- bytes are in the hot path of every connection, including a bad mobile link
- if you ship an Electron app tho...

The state of the post-quantum internet

done



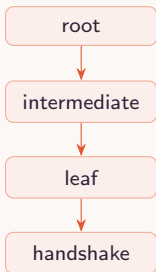
not started

FN-DSA still draft · hybrid certificate format still unsettled · Merkle tree certificates in testing

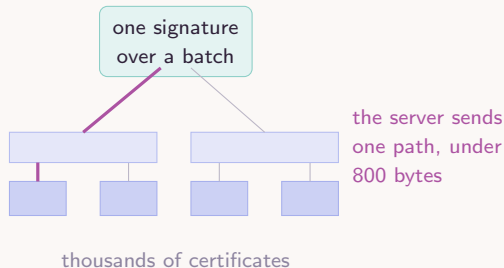
- Encryption is nearly done, took ten years, most middleboxes.
- Authentication is experimental.
- Cloudflare and Tranco scans, mid-2026.

If it can be broken, it can be fixed.

Merkle Tree Certificates



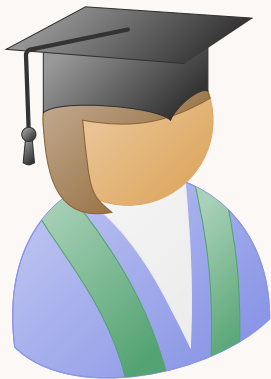
five signatures, sent every time



- PQ signatures are large
- Sign batch once, proves membership: four of the five signatures disappear.
- trust moves from “verify a chain now” to “I already know the root”.
- PQ+auth could be faster than classical

When does a useful quantum computer arrive?

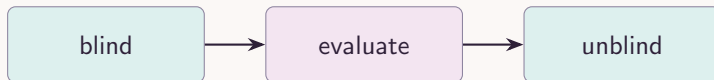
- Estimates from the early 2030s to the mid-2040s
- A lot of arguments: Hybrids, key sizes, which curve to keep
- Transition is as soon as 2029 (Google, Cloudflare)



the Academic

“What is to be done?”

Privacy gets more difficult



$H(x)^r$, then $(\cdot)^k$, then take the r -th root

- Exact, leak-free blinding: Kuperberg.
- Need to slap on a lot of zero-knowledge proofs and counting arguments.

- **HAWK fell** to a classical attack.

It used structure the other lattices do not have.

- **A quantum algorithm claimed the dihedral coset problem in polynomial time.**

ended up strengthening lattices: conjecture that whole families of quantum lattice attacks may never gain more than a polynomial factor.

- **Distinguisher Classic McEliece**, became a quasi-polynomial key recovery.

The great human error is to reason in place of finding out.



We choose without knowing, may be the only kind of choosing there is.

I am afraid we will lose privacy. The protocols are young, and we odn't know how to port all.

Math is beautiful, we get to open systems nobody has opened in thirty years, and not everything is decided.

Stop being sad.

Break something. On a laptop. Over a weekend.

Or fix something.

References

- Sophie Schmieg, *A very unscientific guide to the security of various PQC algorithms*.
<https://keymaterial.net/2025/12/13/a-very-unscientific-guide-to-the-security-of-various-pqc-algorithms/>
- Security Cryptography Whatever, *AI lattice proofs, with Chris Peikert*.
<https://securitycryptographywhatever.com/2026/08/26/ai-lattice-proofs-with-chris-peikert/>
- Bas Westerbaan, *State of the post-quantum Internet*, and the rest of the Cloudflare research blog.
<https://bas.westerbaan.name>
- Sam Jaques, *Landscape of quantum computing*.
https://sam-jaques.appspot.com/quantum_landscape

Thank you.

heimberger.xyz